

Rahul Kumar Ghosh

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🔗 Codeforces 🔗 LeetCode 🌐 Portfolio

Education

Northern University of Business and Technology Khulna

Khulna, Bangladesh

B.Sc. in Computer Science & Engineering

2022 – 2026

CGPA: **3.884** / 4.00 | 152 Credit Hours

Relevant Coursework: Data Structures, Algorithms Design & Analysis, Artificial Intelligence, Neural Networks, Pattern Recognition, Compiler Design, Operating Systems, Database Management Systems, Computer Architecture, Theory of Computation, Software Engineering, Computer Networks, Discrete Mathematics, Mathematical Analysis for Computer Science, Linear Programming & Combinatorial Optimization.

Research Interests

Large Language Models, Deep Learning, Low Rank Adaptation (LoRA), Quantum Error Correction, Bridge between Classical and Quantum Computing, Medical Image Processing, Transformers Competitive Programming, Algorithms, Machine Learning, Explainable AI (XAI), Natural Language Processing, Human Computer Interaction(HCI)

Competitive Programming

Contest		Rank	Scope
BUET Inter University Programming Contest 2026	Team: <i>NUBTK \$(^w^)\$</i>	88	National
ICPC Dhaka Regional 2025	Team: <i>NUBTK \$(^w^)\$</i>	83	National
ICPC Dhaka Regional 2025 — Preliminary	Team: <i>NUBTK \$(^w^)\$</i>	45	National
NWU IUPC 2025	Team: <i>NUBTK \$(^w^)\$</i>	23	National
UU IUPC 2025	Team: <i>trio_QUbit</i>	101	National
DUET IUPC 2025	Team: <i>trio_QUbit</i>	82	National
KUET IUPC 2025	Team: <i>trio_QUbit</i>	140	National
ICPC Dhaka Regional 2024 — Preliminary	Team: <i>trio_QUbit</i>	391	National
KU KRIUPC 2024	Team: <i>trio_QUbit</i>	37	Divisional

Problem Setting

- [Intra University Programming Contest 2025 — CSE Fest Sfuron 22](#)
- [Intra University Programming Contest 2025](#)
- [Inter University Online Contest 2024 \(Div 2\)](#)
- [Inter University Online Contest 2024 \(Div 1+2\)](#)

Projects

Application Generator — A web-based formal university application generator for automating structured academic requests.

Cover Page Generator — An automated NUBTK assignment cover page generator with customizable fields and PDF export.

10000 Hour — A deliberate practice tracker designed to help users master any skill through structured 10,000-hour milestones.

Word Chain — An interactive word chain game built with vanilla JavaScript.

Playlist — A personal music playlist collection and player web application.

Technical Skills

Languages	C, C++, Python, Java, Rust, Assembly(x86, ARM) JavaScript, HTML/CSS, $\text{\LaTeX}$
AI/ML	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
Tools	Git, GitHub, Linux (Arch, Mint), VS Code, Jupyter
Domains	Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Explainable AI

Interests

- Mathematics (Number Theory, Linear Algebra, Calculus)
- Physics (Quantum Mechanics, Electromagnetism)
- Quantum Computing
- Linux (Arch Linux, Linux Mint)
- Competitive Programming
- Combat Sports (MMA, Brazilian Jiu-Jitsu)

Online Profiles

GitHub	github.com/QKnot
LinkedIn	linkedin.com/in/qknot
Codeforces	codeforces.com/profile/qknot
LeetCode	leetcode.com/QKnot
Portfolio	qknot.github.io/Portfolio